

Land Survey

Engineering Mathematics III

Learner Notes

Survey data analysis, matrices, probability, statistics, errors, and adjustment calculations.

Learning outcome	Focus
Analyze survey data	Applied mathematics in workplace and training contexts.
Apply matrices and statistics	Applied mathematics in workplace and training contexts.
Interpret errors and accuracy	Applied mathematics in workplace and training contexts.

1. Topic Overview

This topic develops mathematical skills required in Land Survey. Learners should show clear method, correct units, and technical interpretation.

2. Key Formulas

$$\bar{x} = \frac{\sum x}{n}$$

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$AX = B$$

3. Worked-Example Method

Step 1: identify the given data. Step 2: select the correct formula. Step 3: substitute values carefully. Step 4: compute. Step 5: interpret the answer with correct units.

$$\bar{x} = \frac{\sum x}{n}$$

4. Practice Questions

1. Calculate mean and standard deviation of repeated observations.

2. Solve a matrix system from survey adjustment.
3. Compute probability from survey quality data.